Math 4261: Mathematical Statistics I

Lecture 14: Fisher information and Cramer-Rao
(Reference: Rice 8.5)

Fall 2026

Outline

Asymptotic normality and confidence intervals

Fisher information

Cramer-Rao lower bound

Fisher information for multiple parameters

Recap: Theoretical properties of MLE

Suppose we have i.i.d. observations $X_1, \ldots, X_n$ of $X \sim f(x \mid \theta)$. Then under suitabel conditions, the MLE $\widehat{\theta}$ satisfies

- **Consistency** $\widehat{\theta} \to \theta$ in probability.
- **Asymptotic normality** $\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, \frac{1}{I(\theta)})$.

The asymptotic variance depends on the Fisher information

$$I(\theta) = \mathrm{Var}_\theta \left[\frac{\partial}{\partial \theta} \log f(X \mid \theta) \right] = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f(X \mid \theta) \right].$$

Recap: Proof of MLE consistency

Suppose $X \sim f(x \mid \theta_*)$, so θ_* is the true unknown parameter. Three ingredients:

- Optimizing $\ell_n(\theta)$ in the MLE procedure is, with high probability, the same as optimizing $L(\theta)$, where by the LLN,

$$\frac{1}{n}\ell_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log f(X_i \mid \theta) \xrightarrow{p} L(\theta) = \mathbb{E}_{\theta_*}[\log f(X \mid \theta)].$$

- The function $L(\theta)$ is maximized at the true parameter θ_*.
- Under suitable conditions, the value of θ maximizing $\frac{1}{n}\ell_n(\theta)$, which is the MLE $\widehat{\theta}$, converges in probability to the value of θ maximizing the limiting function $L(\theta) = \mathbb{E}_{\theta_*}[\log f(X \mid \theta)]$, which is θ_*.

Recap: Proof of MLE asymptotic normality

A Taylor expansion of $\ell_n'(\widehat{\theta})$ around θ gave

$$\sqrt{n}(\widehat{\theta} - \theta) \approx \frac{\frac{1}{\sqrt{n}}\ell_n'(\theta)}{-\frac{1}{n}\ell_n''(\theta)}.$$

- By CLT: $\frac{1}{\sqrt{n}}\ell_n'(\theta) \xrightarrow{d} \mathcal{N}(0, I(\theta))$.
- By LLN: $-\frac{1}{n}\ell_n''(\tilde{\theta}) \xrightarrow{p} I(\theta)$.
- Combine with Slutsky's lemma to get

$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{I(\theta)}\right).$$

Recap: Plug-in estimators

Suppose instead we want to estimate $g(\theta)$.

The **plug-in** estimator, based on the MLE $\widehat{\theta}$, is $g(\widehat{\theta})$.

By the Delta method, for nice enough g,

$$\sqrt{n}(g(\widehat{\theta}) - g(\theta)) \xrightarrow{d} \mathcal{N}\left(0, \frac{g'(\theta)^2}{I(\theta)}\right).$$

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Confidence intervals

Given a coverage level $1-\alpha \in (0,1)$, a $(1-\alpha)$-confidence interval for θ is an interval $\widehat{I} = \widehat{I}(X_1, \ldots, X_n)$ such that

$$\mathbb{P}_\theta[\theta \in \widehat{I}] = 1 - \alpha$$

where $\mathbb{P}_\theta$ is the probability under $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$.

Here θ is a fixed parameter and $\mathbb{P}_\theta$ is the probability over the randomness of the data $X_1, \ldots, X_n$ defining $\widehat{I}$.

The interval $\widehat{I}$ is an asymptotic $(1-\alpha)$-confidence interval for θ if

$$\mathbb{P}_\theta[\theta \in \widehat{I}] \to 1 - \alpha, \quad \text{as } n \to \infty.$$

Informally, the probability that $\widehat{I}$ **covers** θ is approximately $1-\alpha$ for large sample sizes n.

Confidence intervals from asymptotic normality

Proposition

Suppose $\widehat{\theta}$ is a consistent and asymptotically normal estimator of θ, with

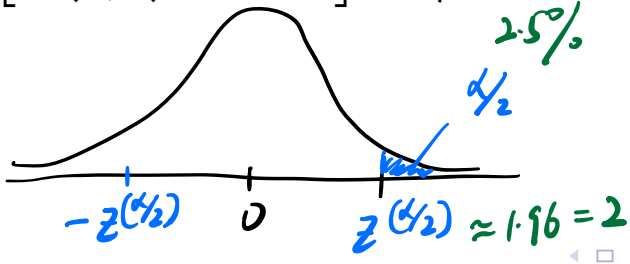
$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, v(\theta)),$$

where $v(\theta)$ is a continuous function and non-zero at θ. Then for any coverage level $1 - \alpha \in (0, 1)$,

$$\widehat{\theta} \pm z^{(\alpha/2)} \sqrt{\frac{v(\widehat{\theta})}{n}}$$

is an asymptotic $(1 - \alpha)$-confidence interval for θ.

This number $z^{(\alpha/2)}$ is the upper-$\alpha/2$ point of the standard normal distribution i.e. $\mathcal{P}\big[\mathcal{N}(0, 1) \geq z^{(\alpha/2)}\big] = \alpha/2$:



Confidence intervals from asymptotic normality

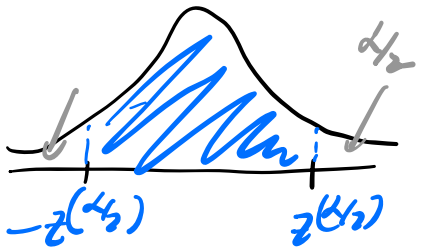
Proof: $P_\theta\left[\theta \in \hat{\theta} \pm z^{(\alpha/2)}\sqrt{\frac{V(\hat{\theta})}{n}}\right]$

$= P_\theta\left[\hat{\theta} - \theta \in \pm z^{(\alpha/2)}\sqrt{\frac{V(\hat{\theta})}{n}}\right]$ $\leftarrow [-z^{(\alpha/2)}, z^{(\alpha/2)}]$

$= P_\theta\left[\sqrt{\frac{n}{V(\hat{\theta})}}(\hat{\theta} - \theta) \in \pm z^{(\alpha/2)}\right]$ (*)

- Asymp. normality: $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, V(\theta))$
- $\hat{\theta} \to \theta$ in prob. by consistency
  so $V(\hat{\theta}) \to V(\theta)$ in prob. by cont. mapping theorem
- Slutsky's $\sqrt{\frac{n}{V(\hat{\theta})}}(\hat{\theta} - \theta) \to N(0,1)$

So (*) $\to 1 - \alpha$



Example

Let $X_1, \ldots, X_n$ be i.i.d. Pois(λ). Find an asymptotic $(1-\alpha)$-CI for λ.

Recall MLE $\hat{\lambda} = \bar{X}$,

Fisher info $I(\lambda) = \frac{1}{\lambda}$.

$V(\lambda) = \lambda$

Asymptotic normality of MLE: $\sqrt{n}(\hat{\lambda} - \lambda) \xrightarrow{d} N(0, \lambda)$

An asymptotic $(1-\alpha)$-CI for λ is

$$\left[\bar{X} - z^{(\alpha/2)} \sqrt{\frac{\bar{X}}{n}}, \; \bar{X} + z^{(\alpha/2)} \sqrt{\frac{\bar{X}}{n}} \right]$$

For 95%-CI, $\alpha = 0.95$, $\frac{\alpha}{2} = 0.05$, put $z^{(\alpha/2)} \approx 1.96$

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Proof sketch: Definition of Fisher information

Let us now check the equivalence of the definitions

$$I(\theta) = \mathrm{Var}_\theta\left[\frac{\partial}{\partial\theta}\log f(X \mid \theta)\right] = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2}\log f(X \mid \theta)\right],$$

and that the score has mean zero.

① Show $\mathbb{E}_\theta\left[\frac{\partial}{\partial\theta}\log f(x|\theta)\right] = 0$

Note $\int f(x|\theta)\,dx = 1 \Rightarrow 0 = \frac{\partial}{\partial\theta}\int f(x|\theta)\,dx$

$= \int \frac{\partial}{\partial\theta} f(x|\theta)\,dx$

Then

$$\mathbb{E}_\theta\left[\frac{\partial}{\partial\theta}\log f(x|\theta)\right] = \mathbb{E}_\theta\left[\frac{\frac{\partial}{\partial\theta}f(x|\theta)}{f(x|\theta)}\right] \overset{\text{LOTUS}}{=} \int \frac{\frac{\partial}{\partial\theta}f(x|\theta)}{f(x|\theta)} f(x|\theta)\,dx = 0$$

Proof sketch: Definition of Fisher information

(2) Differentiating again in θ

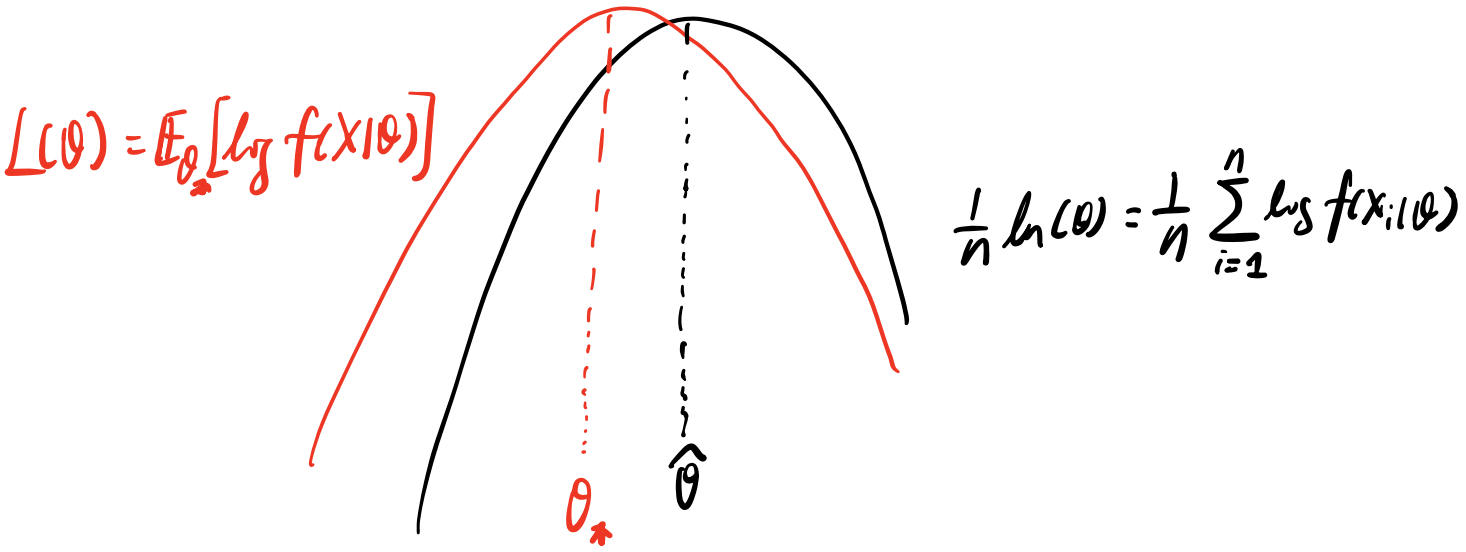
$$0 = \frac{\partial}{\partial\theta}\mathbb{E}_\theta\left[\frac{\partial}{\partial\theta}\log f(x|\theta)\right]$$

$$= \int \frac{\partial}{\partial\theta}\left\{\left[\frac{\partial}{\partial\theta}\log f(x|\theta)\right]\cdot f(x|\theta)\right\}dx$$

$$= \int\left[\frac{\partial^2}{\partial\theta^2}\log f(x|\theta)\right]f(x|\theta) + \int\left[\frac{\partial}{\partial\theta}\log f(x|\theta)\right]\underbrace{\frac{\partial}{\partial\theta}f(x|\theta)}_{=\left[\frac{\partial}{\partial\theta}\log f(x|\theta)\right]\cdot f(x|\theta)}$$

$$= \mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2}\log f(x|\theta)\right] + \underbrace{\mathbb{E}_\theta\left[\left(\frac{\partial}{\partial\theta}\log f(x|\theta)\right)^2\right]}_{=\mathrm{Var}_\theta\left[\frac{\partial}{\partial\theta}\log f(x|\theta)\right]\ \text{b/c mean zero.}}$$

Expected log-likelihood function

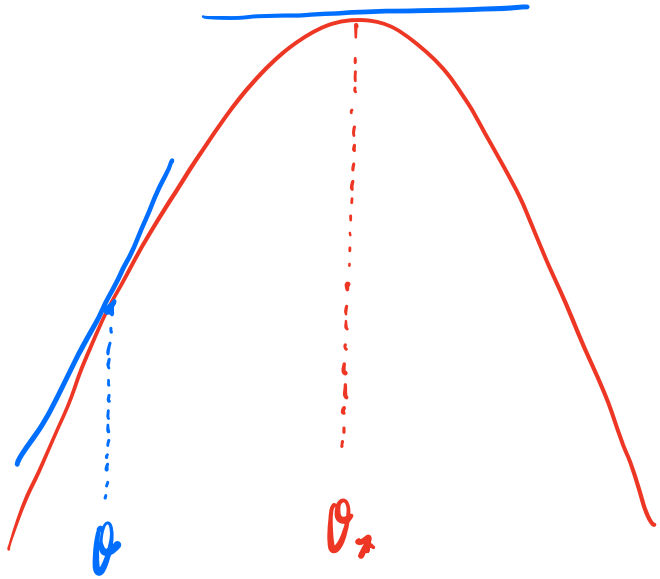


(Recap) If the true parameter is θ_*, then by the LLN, the average log-likelihood approaches the expected log-likelihood

$$\frac{1}{n}\ell_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log f(X_i \mid \theta) \to L(\theta) = \mathbb{E}_{\theta_*}[\log f(X \mid \theta)]$$

in probability. The expected log-likelihood $L(\theta)$ is maximized at the true parameter θ_*.

Expected score

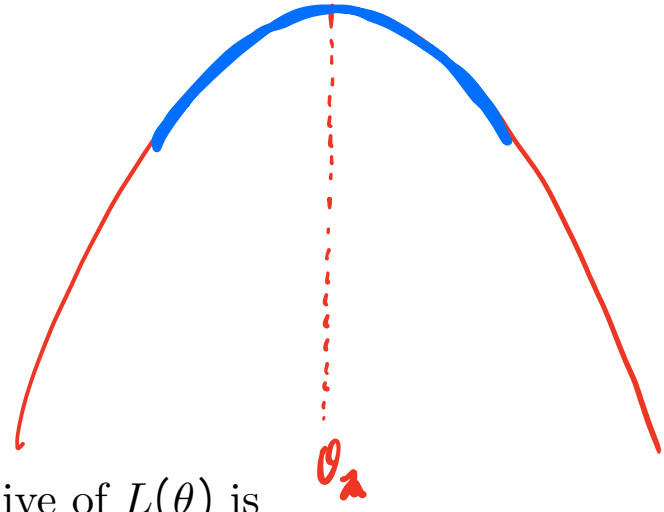


The derivative of $L(\theta)$ is

$$L'(\theta) = \frac{\partial}{\partial\theta}\mathbb{E}_{\theta_*}[\log f(X \mid \theta)] = \mathbb{E}_{\theta_*}[\underbrace{\frac{\partial}{\partial\theta}\log f(X \mid \theta)}_{\text{score}}]$$

So $L'(\theta)$ is the expected score. Since $L(\theta)$ is maximized at θ_*, this also explains why the expected score is zero at the true parameter θ_*.

Geometry of Fisher information



The second derivative of $L(\theta)$ is

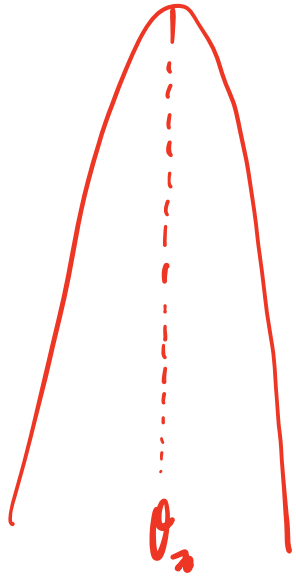
$$L''(\theta) = \frac{\partial^2}{\partial\theta^2}\mathbb{E}_{\theta_*}[\log f(X \mid \theta)] = \mathbb{E}_{\theta_*}\left[\frac{\partial^2}{\partial\theta^2}\log f(X \mid \theta)\right]$$

In particular, at the **true parameter** θ_*,

$$-L''(\theta_*) = -\mathbb{E}_{\theta_*}\left[\frac{\partial^2}{\partial\theta^2}\log f(X \mid \theta)\Big|_{\theta=\theta_*}\right] = I(\theta_*)$$

So $I(\theta_*)$ measures the curvature of $L(\theta)$ around its maximizer θ_*.

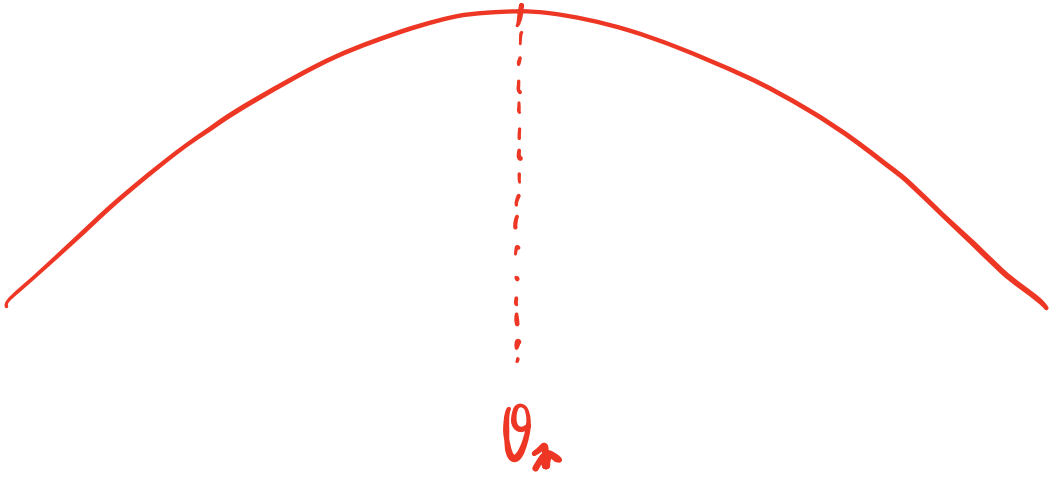
Large Fisher information



When Fisher information $I(\theta_*) = -L''(\theta_*)$ is **large**, moving θ slightly away from θ_* leads to a large decrease in the log-likelihood $L(\theta)$.

That is, a small change in θ makes the observed data much less probable. In this case, the data contains more "information" that the variable θ is close to the true parameter θ_*.

Small Fisher information



Conversely, when Fisher information $I(\theta_*) = -L''(\theta_*)$ is **small**, small changes in θ near θ_* only decrease the log-likelihood $L(\theta)$ by a small amount.

That is, the observed data remains highly probable over a wider neighborhood of θ_*. In this case, the data contains less "information" that the variable θ is close to the true parameter θ_*.

Fisher information as an intrinsic property

To summarize

- $I(\theta)$ represents the curvature of the expected log-likelihood around its maximum, when θ is the true parameter.
- $I(\theta)$ quantifies the rate of decrease of the expected log-likelihood around the true parameter. It can be interpreted as a measure of the expected amount of information the data contains about θ.

We have shown that $I(\theta)$ is related to the asymptotic variance of the MLE. But we will see later today that it may also be understood as a **fundamental quantity** of the model $f(x \mid \theta)$ without reference to any specific estimation procedure.

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Cramer-Rao lower bound

Theorem

Under regularity conditions, let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$ where θ is a single parameter. Let $T = T(X_1, \ldots, X_n)$ be any unbiased estimator of θ. Then

$$\text{Var}_\theta[T] \geq \frac{1}{nI(\theta)}$$

This is an example of an *information-theoretic* lower bound, stating that $\frac{1}{nI(\theta)}$ is a fundamental limit to how accurate any unbiased estimator of θ can be[1].

Modern interpretation: Despite any computing power we can throw at the problem, this is the best we can do.

[1] Under additional regularity conditions, there are also results that show that no estimator—unbiased or not—can achieve a lower asymptotic MSE than $\frac{1}{nI(\theta)}$, except possibly on a set of measure zero special values for θ. Not covered in this course.

Proof—Cramer-Rao

Define $Z = \frac{\partial}{\partial\theta} \ln L(\theta) = \sum_{i=1}^{n} \underbrace{\frac{\partial}{\partial\theta} \log f(X_i|\theta)}_{\text{Score at } X_i}$

Recall, at true parameter θ,

$$\mathbb{E}_\theta\left[\frac{\partial}{\partial\theta} \log f(X|\theta)\right] = 0, \quad \mathrm{Var}_\theta\left[\frac{\partial}{\partial\theta} \log f(X|\theta)\right] = I(\theta)$$

$$\Rightarrow \mathbb{E}_\theta Z = 0, \quad \mathrm{Var}_\theta Z = n I(\theta).$$

Correlation between Z and any unbiased est. T is

$$\frac{\mathrm{Cov}_\theta(Z, T)}{\sqrt{\mathrm{Var}_\theta Z}\sqrt{\mathrm{Var}_\theta T}} \in [-1, 1]$$

Proof—Cramer-Rao

$$\text{So} \quad \frac{\text{Cov}_\theta^2(Z, T)}{\text{Var}_\theta Z \cdot \text{Var}_\theta T} \leq 1$$

$$\Rightarrow \text{Var}_\theta T \geq \frac{\text{Cov}_\theta^2[Z, T]}{\text{Var}_\theta Z} = \frac{\text{Cov}_\theta^2[Z, T]}{n I(\theta)}$$

T is an unbiased estimator of θ. So for all θ,

$$\theta = \mathbb{E}_\theta T \overset{\text{LOTUS}}{=} \int T(x_1, \ldots x_n) f(x_1, \ldots x_n | \theta) \, dx_1 \ldots dx_n$$

Proof—Cramer-Rao

Differentiate in θ

$$\left(\frac{\partial}{\partial\theta}\log f(x_1 \ldots x_n|\theta)\right)\cdot f(x_1 \ldots x_n|\theta)$$

$$1 = \int T(x_1 \ldots x_n)\,\frac{\partial}{\partial\theta} f(x_1 \ldots x_n|\theta)\,dx_1 \ldots dx_n$$

$$= \mathbb{E}_\theta\left[T(X_1 \ldots X_n)\cdot \frac{\partial}{\partial\theta}\log f(X_1 \ldots X_n|\theta)\right]$$

$$= \sum_{i=1}^{n} \frac{\partial}{\partial\theta}\log f(X_i|\theta) = Z$$

$$\Rightarrow \quad 1 = \mathbb{E}_\theta[TZ]$$

$$\text{So} \quad \mathrm{Cov}_\theta[T, Z] = \mathbb{E}_\theta(TZ) - \mathbb{E}_\theta[T]\,\underbrace{\mathbb{E}_\theta[Z]}_{=0} = 1$$

$$\text{Finally} \quad \mathrm{Var}_\theta[T] \geq \frac{1}{n\,I(\theta)}$$

Asymptotic efficiency

[next class]

An estimator $\widehat{\theta}$ is **asymptotically efficient** if

$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{I(\theta)}\right).$$

That is, the variance of the Normal approximation to $\widehat{\theta}$ for large n attains the Cramer-Rao lower bound of $\frac{1}{nI(\theta)}$. Our main theorem thus shows that the MLE is asymptotically efficient.

If two estimators $\widehat{\theta}_1$ and $\widehat{\theta}_2$ based on $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$ satisfy

$$\sqrt{n}(\widehat{\theta}_1 - \theta) \xrightarrow{d} \mathcal{N}(0, v_1(\theta))$$
$$\sqrt{n}(\widehat{\theta}_2 - \theta) \xrightarrow{d} \mathcal{N}(0, v_2(\theta))$$

where $v_2(\theta) < v_1(\theta)$, then $v_2(\theta)/v_1(\theta)$ is the **asymptotic relative efficiency** of $\widehat{\theta}_1$ relative to $\widehat{\theta}_2$.

Asymptotic relative efficiency

Since the Normal approximations have variances $\frac{v_1(\theta)}{n}$ and $\frac{v_2(\theta)}{n}$ for large n, relative efficiency may also be interpreted as the **ratio of sample sizes** needed for $\widehat{\theta}_1$ and $\widehat{\theta}_2$ to achieve the same variance:

If $v_2(\theta) = 80\% \times v_1(\theta)$, then $\widehat{\theta}_1$ using n samples achieves roughly the same variance as $\widehat{\theta}_2$ using only $80\% \times n$ samples.

Example: (Revisited). For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Pareto}(\theta, 1)$, for $\theta > 2$, we showed that

$$\sqrt{n}(\widehat{\theta}_{\text{MoM}} - \theta) \xrightarrow{d} \mathcal{N}\left(0, \frac{\theta(\theta-1)^2}{\theta-2}\right)$$

$$\sqrt{n}(\widehat{\theta}_{\text{MLE}} - \theta) \xrightarrow{d} \mathcal{N}\left(0, \theta^2\right)$$

Thus, the asymptotic relative efficiency of the MoM to MLE is $\frac{\theta(\theta-2)}{(\theta-1)^2}$ which is < 1.

Cramer-Rao for plug-in estimators

The Cramer-Rao lower bound also holds for unbiased estimators of $g(\theta)$.

For the MLE $\widehat{\theta}$, the plug-in estimate $g(\widehat{\theta})$ for $g(\theta)$ satisfies, by the delta method,

$$\sqrt{n}(g(\widehat{\theta}) - g(\theta)) \xrightarrow{d} \mathcal{N}\left(0, \frac{g'(\theta)^2}{I(\theta)}\right).$$

Thus $\operatorname{Var}\left[g(\widehat{\theta})\right] \approx \frac{g'(\theta)^2}{nI(\theta)}$ for large n.

Theorem

Under regularity conditions, let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$ where θ is a single parameter. Let $T = T(X_1, \ldots, X_n)$ be any unbiased estimator of $g(\theta)$. Then

$$\operatorname{Var}_\theta[T] \geq \frac{g'(\theta)^2}{nI(\theta)}.$$

Bottom line: efficiency of MLE carries over to smooth functions.

Example: Plug-in estimator for Pareto

Recall that for Pareto$(\theta, 1)$, we found the MLE $\widehat{\theta} = \frac{n}{\sum_{i=1}^{n} \log X_i}$.

The plug-in estimate $\frac{\widehat{\theta}}{\widehat{\theta}-1}$ is not the only reasonable estimate of the mean of the Pareto distribution. What if we decided to simply use the **sample mean** $\overline{X}$?

For this estimate $\overline{X}$, the CLT gives

$$\sqrt{n}\left(\overline{X} - \frac{\theta}{\theta - 1}\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{\theta}{(\theta-1)^2(\theta-2)}\right)$$

where $\frac{\theta}{(\theta-1)^2(\theta-2)}$ is the variance of the Pareto distribution assuming $\theta > 2$.

Example: Plug-in estimator for Pareto

For $\theta > 2$, comparing the plug-in and sample mean estimators:

$$\sqrt{n}\left(\frac{\widehat{\theta}}{\widehat{\theta}-1} - \frac{\theta}{\theta-1}\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{\theta^2}{(\theta-1)^4}\right)$$

$$\sqrt{n}\left(\overline{X} - \frac{\theta}{\theta-1}\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{\theta}{(\theta-1)^2(\theta-2)}\right)$$

Check: the asymptotic variance corresponding to $\overline{X}$ is **larger** than that of the plug-in! So for large n, the plug-in estimate using the MLE is more accurate than the sample mean $\overline{X}$.

Intuition: Pareto distribution is *heavy-tailed*, and the sample mean $\overline{X}$ is heavily influenced by rare but large data values. In contrast, the MLE $\widehat{\theta}$ estimates θ using $\log X_i$, and then estimates the mean from its relationship to θ.

Potential downside: The plug-in approach relies strongly on the correctness of the Pareto model. On the other hand $\overline{X}$ would be a reasonable estimate of the mean even if Pareto model isn't truly appropriate.

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~~Fisher information for multiple parameters~~ [next time]